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LLaDA-Image: Building Strong Image Generators with Fully Open Training Recipes

A 6B Diffusion Transformer with Staged Training, Parameter-Free RMSNorm, and the Muon Optimizer Achieves State-of-the-Art on Qwen-Image-Bench with Full Open Release

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High-quality text-to-image generation has long been dominated by closed proprietary systems. LLaDA-Image breaks this pattern: a 6B Diffusion Transformer (DiT) trained from scratch within a unified framework alongside a frozen vision-language understanding module, achieving state-of-the-art scores of 53.53 (English) and 53.38 (Chinese) on Qwen-Image-Bench, with all model weights, training code, and stage-by-stage recipes publicly released. The key engineering decisions — staged training (image-only first, then paired text-image), parameter-free RMSNorm, and the Muon optimizer — are each responsible for significant quality and stability gains and are now fully reproducible by the community.

AT A GLANCE

Problem: Competitive text-to-image generation requires proprietary recipes, closed weights, and unreported training decisions.

Why it matters: Without reproducible open baselines, the community cannot build on, audit, or improve SOTA image generators.

Method: 6B DiT + frozen LLaDA2.0-Mini backbone; image-only pre-training then paired text-image training; parameter-free RMSNorm; Muon optimizer.

Result: SOTA Qwen-Image-Bench 53.53 EN / 53.38 ZH; Muon reduces training compute by 20%; Turbo variant in 2–4 steps.

Evidence: Qwen-Image-Bench, GenEval, T2I-CompBench; ablations on staged training, RMSNorm, and optimizer.

Architecture: Unified DiT Plus Frozen LLM

LLaDA-Image pairs two modules:

Image Generation Module (DiT). A 6-billion-parameter Diffusion Transformer trained from scratch to perform masked discrete diffusion over image tokens. The DiT models the conditional distribution $p(x_{\text{image}}|c_{\text{text}})$ where c_{text} is the conditioning signal from the understanding module.

Vision-Language Understanding Module. A frozen LLaDA2.0-Mini diffusion language model, which processes text prompts and provides conditioning for the DiT. Because LLaDA2.0-Mini was trained on both English and Chinese text, the system natively understands Chinese generation instructions — a key advantage over English-dominant baselines.

The two modules share an image tokeniser (VQ-VAE) that discretises images into sequences of tokens, enabling a unified token vocabulary.

Key Engineering Decision 1: Staged Training

Training starts with image-only pre-training before any text-image pair is introduced.

Stage 1 — Image-only pre-training (98M samples): The DiT is trained on unlabeled images using the masked diffusion objective with no conditioning signal. This establishes a strong visual generative prior: the model learns the manifold of natural images before it is asked to condition on language.

Stage 2 — Paired text-image mid-training (122M samples): The DiT is fine-tuned on (text, image) pairs with conditioning from the understanding module. Starting from a strong visual prior produces significantly better text-following than training on pairs from scratch.

Stage 3 — Instruction fine-tuning: A smaller high-quality stage sharpens precision on detailed captions and editing instructions.

The ablation is decisive: removing Stage 1 and starting with paired training reduces Qwen-Image-Bench score by 3.8 points.

Key Engineering Decision 2: Parameter-Free RMSNorm

Standard affine RMSNorm applies learned scale γ and bias β:

$$\text{RMSNorm}_{\text{affine}}(x) = \gamma \cdot \frac{x}{\|x\|_2} + \beta$$

At 6B parameters, the learned scale can drift during training, causing loss spikes that require manual intervention.

Parameter-free RMSNorm simply normalises to unit norm:

$$\text{RMSNorm}_{\text{free}}(x) = \frac{x}{\|x\|_2}$$

Fixing the output norm eliminates the scale drift channel, enabling stable training throughout. Applied uniformly throughout the DiT, this removes approximately 12M parameters while improving training stability.

Key Engineering Decision 3: Muon Optimizer

Standard Adam-family optimisers update parameters along the raw gradient direction, which can inflate or collapse parameter norms. The Muon optimizer applies orthogonalised gradient updates:

$$\mathbf{G}_t^{\perp} = \text{NewtonSchulz5}(\mathbf{M}_t)$$

$$\theta_{t+1} = \theta_t - \eta \mathbf{G}_t^{\perp}$$

where $\mathbf{M}_t$ is the momentum-corrected gradient and NewtonSchulz5 applies five iterations of the Newton-Schulz iteration to orthogonalise $\mathbf{M}_t$. Orthogonal updates preserve parameter norms, improving both convergence speed and final quality.

Masked Diffusion Objective

For an image token sequence $x = (x_1, \ldots, x_T)$, the DiT is trained to predict masked tokens:

$$\mathcal{L}_{\text{DiT}} = -\mathbb{E}_{t,\mathbf{m}}\left[\sum_{i:m_i=1} \log p_\theta(x_i | x_{\setminus m}, c, t)\right]$$

where $\mathbf{m}$ is a random mask, $x_{\setminus m}$ are unmasked tokens, c is text conditioning, and t is the diffusion timestep.

LLaDA-Image-Turbo: Distillation to 2–4 Steps

LLaDA-Image-Turbo is distilled from the full model using consistency distillation:

$$\mathcal{L}_{\text{turbo}} = \mathbb{E}_{x,t,s}\left\| f_\theta(x_t, t) - f_{\theta^-}(x_s, s) \right\|^2$$

where $t > s$ are timesteps on the same trajectory, f_θ is the student, and f_{θ^-} is an EMA teacher. Enforcing consistency along generation trajectories enables convergence in 2–4 steps at 94–97% of full model quality.

Experimental Results

Model	QB-EN	QB-ZH	GenEval
DALL-E 3	50.1	48.7	0.61
SD 3.5-L	49.6	—	0.71
Flux-1.1-pro	51.4	—	0.72
LLaDA-Image	**53.53**	**53.38**	**0.74**
LLaDA-Turbo (4-step)	51.8	51.4	0.71

Training compute: Muon reduces GPU-hours to convergence by 20% (75,500 vs. 94,400 with AdamW) and reaches the same validation loss 25% fewer steps.

Ablations

Stage 1 removal: −3.8 Qwen-Image-Bench points.

Affine vs. free RMSNorm: Affine causes training instability at 4B+ scale; free RMSNorm eliminates all spikes.

Muon vs. AdamW: Muon is +1.1 Qwen-Image-Bench points at the same compute budget.

Turbo step count: 4 steps retains 97% of quality; 2 steps 94%, at 8× speedup.

Chinese prompt following: The bilingual LLaDA2.0-Mini backbone provides SOTA Chinese scores where English-dominant baselines degrade significantly.

Reference

Chuyan Chen et al. (inclusionAI). **LLaDA-Image: Building Strong Image Generators with Fully Open Training Recipes.** arXiv:2609.03796, September 2026. https://arxiv.org/abs/2609.03796